

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2. The solution is built on the ServiceNow Now Platform: commuters book tickets in the Service Portal, Flow Designer automates booking, simulated payment, QR generation and notification, and the Gate Validation page (backed by a Scripted REST API) validates tickets at entry and exit.

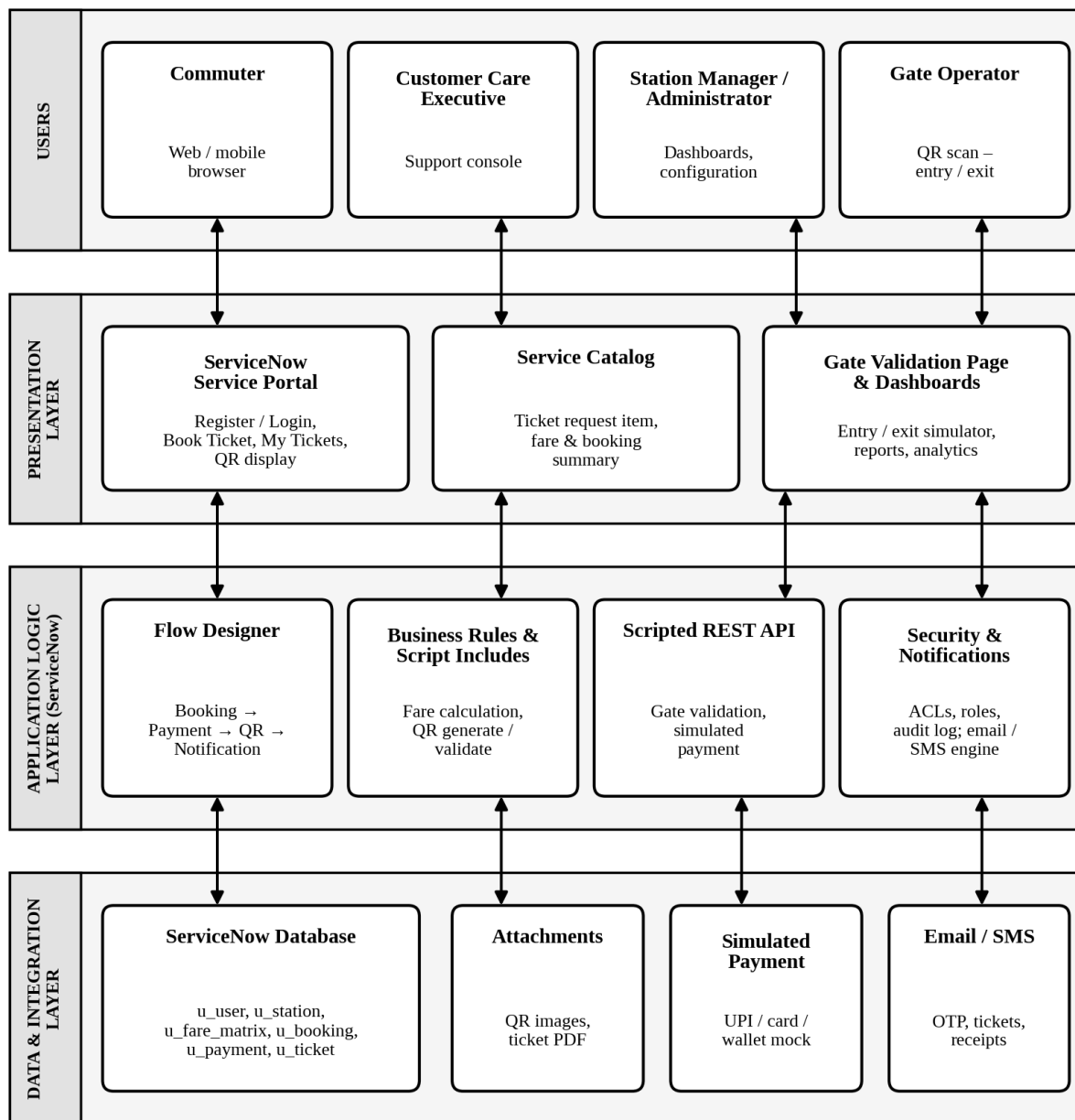


Figure 1: Technical architecture – Metro Ticket Generating System in ServiceNow

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web / mobile-browser interface through which commuters register, select stations, pay and view QR tickets; staff interface for gate validation, support and dashboards.	ServiceNow Service Portal, Service Catalog, HTML, CSS, JavaScript, AngularJS widgets
2.	Application Logic-1	Business logic for fare calculation, booking creation, ticket status rules and validation of user input.	ServiceNow server-side JavaScript (Business Rules, Script Includes, Client Scripts)
3.	Application Logic-2	Workflow automation of the booking journey: booking → simulated payment → QR generation → notification.	ServiceNow Flow Designer / Workflow Studio
4.	Application Logic-3	Unique QR code generation after payment, and QR validation at entry / exit (valid, unused, unexpired).	ServiceNow Script Include with open-source QR code JavaScript library; Scripted REST API
5.	Database	Stores users, stations, fare matrix, bookings, payments and tickets in custom tables with reference fields between them.	ServiceNow relational database (tables u_user, u_station, u_fare_matrix, u_booking, u_payment, u_ticket)
6.	Cloud Database	Managed database service on cloud, with no separate database installation needed.	ServiceNow Cloud (Now Platform)
7.	File Storage	Storage of QR code images and downloadable ticket files.	ServiceNow Attachments (sys_attachment)
8.	External API-1	Payment step, simulated for the student project: UPI, card and wallet options return success / failure and a transaction reference; no real money or card data is handled.	ServiceNow Script Include / Scripted REST API (simulated payment gateway)
9.	External API-2	Delivery of OTP, registration confirmation, ticket and receipt to the commuter.	ServiceNow Notification Engine (Email); SMS gateway (optional)
10.	Machine Learning Model	Not used in the current scope; peak-hour demand forecasting can be added later using booking data.	Not applicable (future: ServiceNow Predictive Intelligence)
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: Not required (accessed through a web browser). Cloud Server Configuration: ServiceNow Personal Developer Instance (PDI) on the Now Platform.	ServiceNow Cloud instance (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source libraries used alongside the ServiceNow platform for the user interface and QR code creation.	AngularJS, Bootstrap (Service Portal), open-source QR code JavaScript library
2.	Security Implementations	Role-based access control and ACLs for each user type, HTTPS / TLS for data in transit, hashed passwords, unique QR tokens to prevent forgery or reuse, audit logging, and no storage of real payment credentials.	ServiceNow Roles & ACLs, HTTPS / TLS, SHA-256, OWASP guidelines
3.	Scalable Architecture	Multi-tier architecture (presentation, application logic, data) on a cloud platform; stations, routes and fares are configuration-driven, so growth needs no code changes.	ServiceNow Now Platform (3-tier architecture)
4.	Availability	Highly available cloud hosting with data-centre redundancy and failover managed by the platform; issued tickets remain accessible after booking.	ServiceNow Cloud (high availability)
5.	Performance	Indexed tables for quick booking and ticket lookups, asynchronous notifications so that booking is not delayed, and lightweight QR tokens for fast gate validation (target under 2 seconds).	Database indexing, async Business Rules, scheduled jobs

References:

<https://developer.servicenow.com>

<https://docs.servicenow.com>

<https://c4model.com/>